



Unified Scenario Study of Fanaroff–Riley Galaxy and Blazar

Jingyi Ma¹ and Junhui Fan^{1,2,3} ¹ Center for Astrophysics, Guangzhou University, Guangzhou 510006, People's Republic of China; fjh@gzhu.edu.cn² Greater Bay Brand Center of the National Astronomical Data Center, Guangzhou 510006, People's Republic of China³ Astronomy Science and Technology Research Laboratory of Department of Education of Guangdong Province, Guangzhou 510006, People's Republic of China

Received 2026 February 9; revised 2026 April 18; accepted 2026 April 27; published 2026 May 25

Abstract

Radio galaxies have two subclasses: Fanaroff–Riley type I (FR I) and type II (FR II). Within the radio-loud active galactic nucleus unification framework, FR I galaxies are considered to be the misaligned counterparts of BL Lacertae (BL Lac) objects, while FR II galaxies correspond to misaligned flat-spectrum radio quasars (FSRQs). In this work, we analyze 2124 sources, including 75 FR I galaxies, 147 FR II galaxies, 1142 BL Lac objects, and 760 FSRQs. After deriving their kinetic jet power ($\log L_{\text{kin}}$), we classify the sample into two classes according to the Bayesian information criterion (BIC) method and the Gaussian mixture model (GMM) method. The results obtained are as follows: (1) The BIC analysis indicates that the sample is optimally divided into two classes. The GMM analysis assigns a crossing point at $\log L_{\text{kin}} = 45.38$. Consequently, the sources are separated into the weak class ($\log L_{\text{kin}} < 45.38$) and the strong class ($\log L_{\text{kin}} > 45.38$). (2) Using the machine learning method, the sample is effectively classified into two populations in the $\log M_{\text{BH}} - \log L_{1.4\text{GHz}}$ plane. (3) The Support Vector Machine (SVM) approach is applied to derive a dividing line in the $\log R_{\text{radio}} - \log L_{\text{kin}}$ plane. The accretion rates of sources above and below the dividing line exhibit a statistically significant difference. (4) By incorporating the core-dominance parameter ($\log R_{\text{radio}}$), we identify three changing-look blazars located near the SVM dividing line in the $\log R_{\text{radio}} - \log L_{\text{kin}}$ plane. The changing-look region is bounded by the two lines $\log L_{\text{kin}} = -0.02 \log R_{\text{radio}} + 44.63$ and $\log L_{\text{kin}} = -0.02 \log R_{\text{radio}} + 45.69$.

Unified Astronomy Thesaurus concepts: Active galactic nuclei (16); Blazars (164); Fanaroff–Riley radio galaxies (526)

Materials only available in the online version of record: machine-readable tables

1. Introduction

Radio galaxies have strong radio emissions and are a relatively bright type of active galactic nuclei (AGNs). B. L. Fanaroff & J. M. Riley (1974) divided radio galaxies into two categories, Fanaroff–Riley type I (FR I) and Fanaroff–Riley type II (FR II), based on their luminosity at 178 MHz. FR I galaxies are low-luminosity sources with $L_{178\text{ MHz}} \leq 2 \times 10^{25} \text{ W} \cdot \text{Hz}^{-1} \cdot \text{Sr}^{-1}$, while FR II galaxies are high-luminosity sources with $L_{178\text{ MHz}} \geq 2 \times 10^{25} \text{ W} \cdot \text{Hz}^{-1} \cdot \text{Sr}^{-1}$. FR I galaxies exhibit an “edge-darkened” morphology, with emission concentrated near the core and fading along the jets, whereas FR II galaxies are characterized by an “edge-brightened” structure. Based on optical emission-line properties, the radio galaxies can be separated into low-excitation radio galaxies (LERGs) and high-excitation radio galaxies (HERGs; R. G. Hine & M. S. Longair 1979; S. Buttiglione et al. 2010; P. N. Best & T. M. Heckman 2012). P. N. Best & T. M. Heckman (2012) suggested that HERGs are fueled at high rates through radiatively efficient standard accretion disks, while LERGs are powered by radiatively inefficient flows at low accretion rates. Radio galaxies with a “standard” geometrically thin disk and high accretion rate are classified as HERGs, whereas those with low accretion rates, holding an advection-dominated accretion flow disk, are classified as LERGs (P. N. Best & T. M. Heckman 2012; T. M. Heckman & P. N. Best 2014). FR I galaxies are usually LERGs, while FR II galaxies can be LERGs or HERGs (R. D. Baldi & A. Capetti 2010; H. Miraghaei & P. N. Best 2017; P. Grandi et al. 2021;

B. Mingo et al. 2022; X.-H. Ye et al. 2026). The difference between FR I and FR II radio galaxies has been a long-standing problem. It may be due to the interaction of the jet stream with the environmental medium (P. J. Wiita 2000) or the intrinsic nuclear properties of the accretion stream and the jet formation process (e.g., S. A. Baum et al. 1995; G. V. Bicknell 1995; C. S. Reynolds et al. 1996; D. Marchesini et al. 2004; M. J. Hardcastle et al. 2007).

Blazars are a subclass of radio-loud AGNs characterized by relativistic jets oriented at very small angles to the line of sight. Observationally, they exhibit rapid variability, high polarization, apparent superluminal motion, core-dominated radio morphology, and strong high-energy emissions (e.g., R. L. Moore & H. S. Stockman 1981; J. H. Fan et al. 1997, 2016, 2021; G. E. Romero et al. 2000; I. Andruchow et al. 2005; A. A. Abdo et al. 2010b; Y. G. Zheng & L. Zhang 2011; H. B. Xiao et al. 2022). Based on the properties of their emission lines, blazars can also be divided into BL Lacertae (BL Lac) objects, which have weak or no emission lines, and flat-spectrum radio quasars (FSRQs), which show strong emission lines. The spectral energy distribution (SED) of blazars generally exhibits a two-peaked structure (C. M. Urry & P. Padovani 1995; G. Fossati et al. 1998; J. H. Fan et al. 2016; J. Fan et al. 2023). The first peak corresponds to synchrotron emission, extending from the far-infrared to the soft X-ray band, while the second peak arises from inverse Compton scattering, spanning the MeV–TeV energy range (M. Sikora et al. 1994; E. Nieppola et al. 2008; A. A. Abdo et al. 2010a). Alternatively, depending on their peak synchrotron frequency, blazars can be classified into three categories: low synchrotron peak (LSP) blazars, intermediate synchrotron peak (ISP) blazars, and high synchrotron



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peak (HSP) blazars (A. A. Abdo et al. 2010a; J. H. Fan et al. 2016; J. H. Yang et al. 2022). The corresponding types for BL Lac objects are called LBLs, IBLs, and HBLs. G. Fossati et al. (1998) found that the synchrotron peak frequency is anticorrelated with the synchrotron peak luminosity, leading to the proposal of “blazar sequence” from LSP FSRQs to HSP HBLs.

G. Matt et al. (2003) reported that some AGNs may undergo temporary spectral state transitions, a phenomenon referred to as changing-look. This phenomenon is sometimes revealed in a temporary change of the source classification. There are various explanations of the changing-look phenomenon. Some changing-look AGNs are characterized by dramatic variations in their optical/UV emission lines and continuum, or X-ray spectra (e.g., S. Bianchi et al. 2005; K. D. Denney et al. 2014; B. J. Shappee et al. 2014). The bulk Lorentz factor of the jet provides an explanation for variability of broad-line emission in changing-look sources (V. Bianchin et al. 2009; J. J. Ruan et al. 2014). G. Chen et al. (2025) found that the jet’s helical structure could play a crucial role in changing-look source Ton 599. Changing-look AGNs provide clues to the interplay between accretion process and jet activity.

Many previous studies have investigated the FR I/FR II dichotomy and its implications for the unification of radio-loud AGNs. C. M. Urry & P. Padovani (1995) found that the luminosity functions of BL Lac objects can be reproduced from FR I galaxies through relativistic beaming. Similarly, FR II galaxies are unified with quasars through obscuration and beaming effect. In their discussion on the FR I/FR II dichotomy, M. J. Ledlow & F. N. Owen (1996) applied radio power and host galaxy magnitude to demonstrate that the dividing line between FR I and FR II clearly separates the two populations. G. Ghisellini & A. Celotti (2001) suggested that the fundamental parameter involving the FR I/FR II dichotomy is the Eddington accretion rate, which reflects the regulation of accretion modes and nuclear outflows. Q. Wu & X. Cao (2008), adopting a standard conversion from host galaxy mass to black hole mass, found that the relation between jet power and black hole mass aligns with the Ledlow–Owen dividing line. This strongly suggests that most FR I galaxies may have accretion modes different from those of FR II galaxies.

The formation of AGN jets is still not fully understood. The morphological division between FR I and FR II galaxies may be related to the mechanisms of jet formation. Jets may originate from the Blandford–Znajek (BZ) mechanism associated with black hole spin (R. D. Blandford & R. L. Znajek 1977) or from the Blandford–Payne (BP) process associated with the accretion disk (R. D. Blandford & D. G. Payne 1982). When both mechanisms contribute, the jet may be powered by a combination of BZ and BP processes, forming a hybrid model (D. L. Meier 2001). P. Ghosh & M. A. Abramowicz (1997) found that in the BZ mechanism the standard accretion disk can exist in two states: radiation pressure dominated (RPD) and gas pressure dominated (GPD). L. Foschini (2011) found that FSRQs are in the RPD state, whereas BL Lac objects are generally in the GPD state.

Furthermore, previous studies have explored the connections between radio galaxies and blazars. Y. Xu et al. (2009) demonstrated that BL Lac objects and quasars are separated by the Ledlow–Owen dividing line in the black hole mass–jet power ($M_{\text{BH}}-Q_{\text{jet}}$) plane with BL Lac objects characterized by systematically lower Eddington ratios ($L_{\text{bol}}/L_{\text{Edd}}$). Moreover,

they detected a strong correlation between jet power and broad-line luminosity. T. Sbarrato et al. (2014) found a strong correlation between broad-line luminosity and radio luminosity, indicating a close link between jet power and accretion rate. X. Fan & Q. Wu (2019) investigated different classes of AGNs and found that they exhibit systematic distributions on the Eddington ratio–jet power plane, suggesting an evolutionary sequence from FR II HERG to FR II LERG and then to FR I. X. Ye et al. (2025) proposed that HBLs, IBLs, and high-power LBLs may be the beamed counterparts of FR 0, FR I, and FR II LERGs, supporting the cooling model as an explanation for the blazar sequence.

However, E. T. Meyer et al. (2011) and M. Keenan et al. (2021) pointed out that the diversity of AGN jets cannot be explained solely by a continuous evolutionary sequence. Instead, a more complex set of physical factors must be considered, including the jet formation and collimation mechanisms. Properties of the accretion flow and the characteristics of the host galaxy, such as the stellar environment and the intergalactic medium, collectively determine the jet power, radiative properties, and evolutionary pathways.

Building on the framework outlined by E. T. Meyer et al. (2011), which delineated approximate ranges of kinetic jet power for different subclasses, we attempt to use statistical methods to confirm the specific boundaries of these classifications. In this work, the Bayesian information criterion (BIC) method is adopted to classify the kinetic jet power of our sample. Simultaneously, the Gaussian mixture model (GMM) is utilized to determine the classification boundary. The structure of this paper is organized as follows: In Section 2, we present the sample and the corresponding calculation approach. In Section 3, the results of kinetic jet power classification are given. The discussions are shown in Section 4. In Section 5, we present our conclusions. Through our work, we adopt the Python *astropy* package (T. P. Robitaille et al. 2013; Astropy Collaboration et al. 2018, 2022) of the Planck15 cosmology (Planck Collaboration et al. 2016), with the cosmological parameters $H_0 = 67.74 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_M = 0.31$, and $\Omega_\Lambda = 0.69$.

2. Sample and Calculation Method

2.1. Sample

The FR galaxy sample is compiled from C. Zhang & J. Fan (2009), consisting of 75 FR I and 147 FR II galaxies. The multiwavelength data are obtained from the NASA/IPAC Extragalactic Database (NED).⁴ In Table 1, Columns (5), (7), and (9) represent the synchrotron peak frequency, the optical luminosity at $2.4 \times 10^{14} \text{ Hz}$, and the luminosity of synchrotron peak frequency, which are calculated by C. Zhang (2010). Column (6) lists the radio luminosity at 1.4 GHz calculated in this work. Column (8) lists the X-ray luminosity at 1 keV calculated by C. Zhang & J. Fan (2009). Column (10) gives the bolometric luminosity calculated following the method described in Section 2.2. Column (11) presents the kinetic jet power derived using the method described in Section 2.3. Column (12) gives the black hole mass estimated as described

⁴ <http://ned.ipac.caltech.edu/>

Table 1
Sample of FR Galaxies

IAU	Source Name	Class	z	$\log \nu_p$ (Hz)	$\log L_R$ (erg s ⁻¹)	$\log L_O$ (erg s ⁻¹)	$\log L_X$ (erg s ⁻¹)	$\log L_p$ (erg s ⁻¹)	$\log L_{\text{bol}}$ (erg s ⁻¹)	$\log L_{\text{kin}}$ (erg s ⁻¹)	$\log M_{\text{BH}}$ ($M_\odot$)	Method	References	R_{radio}
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
0013+790	3C 6.1	II	0.8404	14.15	44.27	...	44.55	44.94	46.15	46.57	9.30	M_{bulge}	WC16	...
0031+391	3C 13	II	1.351	13.09	44.47	...	44.06	45.88	46.94	46.71	...	...	...	...
0033+183	3C 14	II	1.469	...	44.57	...	45.27	...	...	46.78	9.50	Host-B	MD04	...
0034-01	3C 15	II	0.073	...	41.93	44.52	42.15	...	...	44.98	8.44	$M-\sigma$	...	-1.65
0035-02	3C 17	II	0.2197	13.31	43.13	...	...	44.41	45.56	45.80	8.32	$M-\sigma$	...	...
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

Note. This is a sample table of FR galaxies. Column (1): the International Association of Universities (IAU) name. Column (2): the source name. Column (3): the classification; I for FR I, II for FR II. Column (4): the redshift. Column (5): the synchrotron peak frequency ($\log \nu_p$). Column (6): the radio luminosity at 1.4 GHz in units of erg s⁻¹ ($\log L_R$). Column (7): the optical luminosity at 2.4×10^{14} Hz in units of erg s⁻¹ ($\log L_O$). Column (8): the X-ray luminosity at 1 keV in units of erg s⁻¹ ($\log L_X$). Column (9): the luminosity of synchrotron peak frequency in units of erg s⁻¹ ($\log L_p$). Column (10): the bolometric luminosity in units of erg s⁻¹ ($\log L_{\text{bol}}$). Column (11): the kinetic luminosity in units of erg s⁻¹ ($\log L_{\text{kin}}$). Column (12): the logarithm of black hole mass in units of $M_\odot$ ($\log M_{\text{BH}}$). Column (13): the method adopted for estimating the black hole mass. Column (14): the black hole mass data references. CX04: X. Cao & S. Rawlings (2004); GA08: A. W. Graham (2008); LY06: Y. Liu et al. (2006); MA03: A. Merloni et al. (2003); MD04: D. Marchesini et al. (2004); MR04: R. J. McLure et al. (2004); RA15: A. E. Reines & M. Volonteri (2015); WC16: C. Westhues et al. (2016). Column (15): the logarithmic core dominance parameter.

(This table is available in its entirety in machine-readable form in the [online article](#).)

Table 2
Sample of Blazars

Source Name	Class	z	$\log \nu_p$ (Hz)	$\log L_R$ (erg s^{-1})	$\log L_O$ (erg s^{-1})	$\log L_X$ (erg s^{-1})	$\log L_p$ (erg s^{-1})	$\log L_{\text{bol}}$ (erg s^{-1})	$\log L_{\text{kin}}$ (erg s^{-1})	$\log M_{\text{BH}}$ ($M_\odot$)	R_{radio}
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
4FGL J0001.2–0747	IBL	...	14.1	42.28	45.20	44.78	45.30	...	45.22	...	...
4FGL J0001.5+2113	FSRQ	1.106	13.2	42.97	45.90	45.9	46.17	47.20	45.69	7.54	0.75
4FGL J0003.9–1149	LBL	0.86	13.2	43.07	45.46	...	45.90	46.98	45.76	...	...
4FGL J0004.0+0840	HBL	2.057	15.3	42.52	...	...	46.01	47.28	45.38	...	...
4FGL J0004.3+4614	FSRQ	1.81	13.1	43.12	...	45.35	45.93	46.97	45.79	8.36	...
...	...	...	...	...	...	...	...	...	...	...	...

Note. This is a sample table of blazars. Column (1): the source name. Column (2): the classification; LBL for LSP frequency BL Lac, IBL for ISP frequency BL Lac, HBL for HSP frequency BL Lac, FSRQ for the flat-spectrum radio quasar. Column (3): the redshift. Column (4): the synchrotron peak frequency ($\log \nu_p$). Column (5): the radio luminosity at 1.4 GHz in units of erg s^{-1} ($\log L_R$). Column (6): the optical luminosity at 2.43×10^{14} Hz in units of erg s^{-1} ($\log L_O$). Column (7): the X-ray luminosity at 1 keV in units of erg s^{-1} ($\log L_X$). Column (8): the luminosity of synchrotron peak frequency in units of erg s^{-1} ($\log L_p$). Column (9): the bolometric luminosity in units of erg s^{-1} ($\log L_{\text{bol}}$). Column (10): the kinetic luminosity in units of erg s^{-1} ($\log L_{\text{kin}}$). Column (11): the logarithm of black hole mass in units of $M_\odot$ ($\log M_{\text{BH}}$), given by V. S. Paliya et al. (2021). Column (12): the logarithmic core dominance parameter, given by Z. Pei et al. (2020c).

(This table is available in its entirety in machine-readable form in the [online article](#).)

in Section 4.1. Column (15) shows the radio core dominance parameter calculated following Section 2.4.

The blazar sample is adopted from J. H. Yang et al. (2022), consisting of 1142 BL Lac objects and 760 FSRQs. In Table 2, Columns (4)–(8) list the multiwavelength luminosities taken from J. H. Yang et al. (2022). Column (9) gives the bolometric luminosity calculated following the method described in Section 2.2. Column (10) presents the kinetic jet power derived using the method described in Section 2.3. Column (11) lists the black hole mass adopted from V. S. Paliya et al. (2021), while Column (12) gives the core dominance parameter taken from Z. Pei et al. (2020a). The relevant data and the calculation results are listed in Table 2.

2.2. Luminosity Calculation

The luminosities for all sources in the sample are consistently derived using the following method. The luminosity distance is given by P. R. Capelo & P. Natarajan (2007): $d_L = \frac{c}{H_0} \int_1^{1+z} \frac{1}{\sqrt{\Omega_M x^3 + 1 - \Omega_M}} dx$. The observed flux density (F_ν^{ob}) obtained from NED is K -corrected with the formula $F_\nu = F_\nu^{\text{ob}} (1+z)^{(\alpha_\nu-1)}$, where z is the redshift and α_ν is the spectral index ($F_\nu \propto \nu^{-\alpha_\nu}$). The corresponding luminosity at frequency ν can be calculated: $L_\nu = 4\pi d_L^2 \nu F_\nu$, in units of erg s^{-1} .

As described above, the radio band ($\log L_R$), optical band ($\log L_O$), and X-ray band ($\log L_X$) luminosities are compiled from the literature and calculated. The multiband SED is then fitted with a parabolic function of the form $\log(\nu F_\nu) = A(\log \nu)^2 + B \log \nu + C$, where νF_ν is in units of $\text{erg cm}^{-2} \text{s}^{-1}$ and ν is in units of Hz. Based on coefficients A , B , and C , the synchrotron peak frequency can be derived as $\log(\nu_p) = -\frac{B}{2A}$. The bolometric luminosity is calculated via the following integral: $L_{\text{bol}} = 4\pi d_L^2 \times \ln 10 \int 10^{A(\log \nu)^2 + B \log \nu + C} d(\log \nu)$. Similarly for blazar objects, in the paper by J. H. Yang et al. (2022), the SED diagrams of $\log(\nu F_\nu)$ and $\log \nu$ of all sources were plotted, and the SED was fitted with a parabolic function. The synchrotron peak frequency $\log \nu_p$ was determined from the vertex of the fitted parabola.

2.3. Kinetic Jet Power

C. J. Willott et al. (1999) estimated the jet power of FR II galaxies based on the extended radio luminosity at 151 MHz: $Q_{\text{jet}} = 3 \times 10^{38} f_W L_{151, \text{ext}}^{6/7} \text{W}$, and combined most of the uncertainties into a single factor (f_W). L. Birzan et al. (2004) derived the pV energy associated with the cavities. They found a significant correlation between the kinetic power and the 1.4 GHz radio luminosity, with a scatter of 0.7 dex. There is uncertainty in this method ($E_{\text{cav}} = f_{\text{cav}} \times pV$), which estimates the cavity energy from the measured pressure and volume. Additional heating from the radio jet could imply higher values of f_{cav} . Some authors argued that f_{cav} can be as high as 10 (e.g., A. Nusser et al. 2006). T. M. Heckman & P. N. Best (2014) used the analysis from K. W. Cavagnolo et al. (2010), and a best-fitting linear relation between kinetic jet power and the 1.4 GHz radio emission was established:

$$L_{\text{kin}} = 7 \times 10^{43} f_{\text{cav}} (L_{1.4\text{GHz}}/10^{25} \text{W Hz}^{-1})^{0.68} \text{erg s}^{-1}. \quad (1)$$

Here $L_{1.4\text{GHz}}$ refers to the 1.4 GHz radio luminosity. The adopted value $f_{\text{cav}} = 4$ is consistent with the widely recognized balance between AGN heating and radiative cooling in massive clusters (K. W. Cavagnolo et al. 2010; T. M. Heckman & P. N. Best 2014). The equation derived by T. M. Heckman & P. N. Best (2014) is in agreement with that of C. J. Willott et al. (1999). Hereafter in this work, we adopt Equation (1) to calculate the kinetic jet power. For cluster analysis, the same processing is applied to the entire sample. Therefore, varying f_{cav} is not expected to significantly affect the classification results. The results are presented in Tables 1 and 2.

2.4. Core Dominance Parameter

The radio core dominance parameter, R_{radio} , defined as the ratio of core flux density to extended flux density, has been shown in previous studies to serve as an indicator of the beaming effect. It can be expressed as $R_{\text{radio}} = \eta \{ [\Gamma(1 - \beta \cos \theta)]^{-p} + [\Gamma(1 + \beta \cos \theta)]^{-p} \}$, where η is the intrinsic ratio, θ is the viewing angle, β is the bulk velocity, the Lorentz factor is defined as $\Gamma = (1 - \beta^2)^{-1/2}$, and $p = n + \alpha$. Here n depends on the geometry and physical conditions of the jet: $n = 2$ for a continuous jet and $n = 3$ for

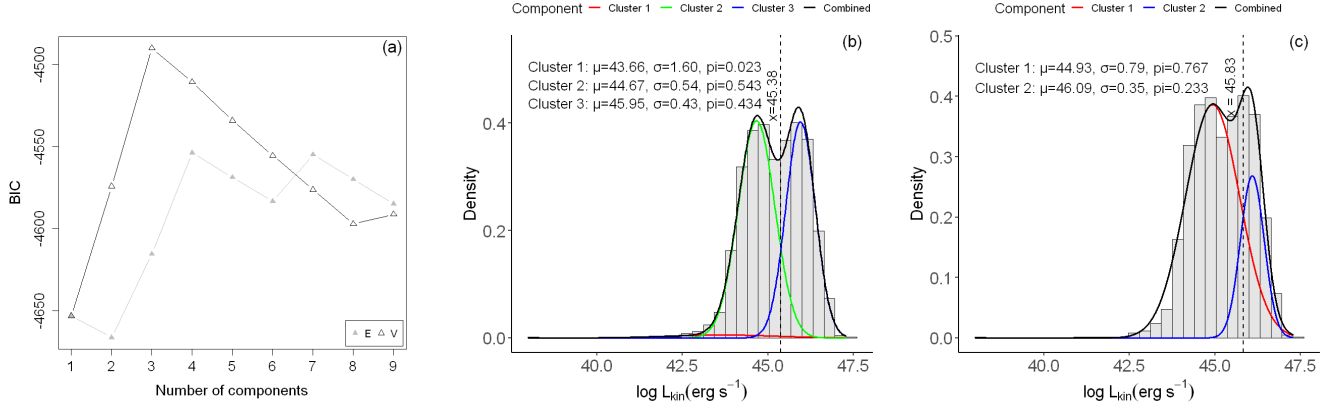


Figure 1. (a) BIC values of GMM fits with different numbers of components. (b) GMM density distribution. Red represents cluster 1, green represents cluster 2, and blue represents cluster 3. The black curve represents the combined fitting result. The gray histogram shows the overall distribution of the data points. (c) GMM density distribution restricted to two Gaussian components.

a moving blob (C. M. Urry & P. Padovani 1995; J. H. Fan 2003).

In order to calculate the radio core dominance parameter and investigate its properties, the radio flux densities used in this work are measured at 5 GHz. Observations at 5 GHz have higher angular resolution and thus provide a more reliable measurement of the core flux (F. K. Liu & Y. H. Zhang 2002). In addition, 5 GHz is a frequency band with relatively abundant available data. Here $F_{\text{core}}^{\text{ob}}$ and $F_{\text{ext}}^{\text{ob}}$ denote the observed flux densities of radio core and extended components, while $F_{\text{core}}^{\text{in}}$ and $F_{\text{ext}}^{\text{in}}$ are the corresponding flux densities after applying the K-correction. This yields the following expressions:

$$F_{\text{core}}^{\text{in}} = F_{\text{core}}^{\text{ob}} (1+z)^{\alpha_{\text{core}}}, \quad F_{\text{ext}}^{\text{in}} = F_{\text{ext}}^{\text{ob}} (1+z)^{\alpha_{\text{ext}}}, \quad (2)$$

where we adopt $\alpha_{\text{core}} = 0.0$ and $\alpha_{\text{ext}} = 0.75$ (J. Fan et al. 2011; Z. Pei et al. 2016, 2020b).

In previous studies, it has been confirmed that the radio core consistently exhibits a flat-spectrum characteristic (e.g., L. Dondi & G. Ghisellini 1995; D. Donato et al. 2001; A. A. Abdo et al. 2010a). A. P. Lobanov (1998) pointed out that in very long baseline interferometry (VLBI) images the core is typically recognized as a compact component characterized by substantial flux and flat spectrum over the radio band. At a given frequency, the core is generally interpreted as the region where the optical depth is $\tau \approx 1$. Synchrotron radiation originates from the motion of relativistic electrons in magnetic fields. At low frequencies, it can be absorbed by the same population of electrons in optically thick regions. This phenomenon is known as synchrotron self-absorption (SSA). Based on the SSA spectrum formulation for AGNs by M. Türlér et al. (1999), the spectral turnover occurs at the peak frequency ν_p^{SSA} , where the optical depth is near unity. The maximum flux, defined by the condition $dF_{\nu}/d\nu = 0$, corresponds to the transition point between the optically thin ($\tau \ll 1$) and optically thick ($\tau \gg 1$) regions. In the model derived by R. D. Blandford & A. Königl (1979), the radio core is identified as the innermost region of the jet where the emission transitions from being optically thick to optically thin. Therefore, the expression for the core dominance

parameter becomes

$$\log R_{\text{radio}} = \log \left(\frac{F_{\text{core}}}{F_{\text{ext}}} \right) (1+z)^{\alpha_{\text{core}} - \alpha_{\text{ext}}}. \quad (3)$$

3. Result

3.1. Kinetic Jet Power Classification

For those 75 FR I galaxies, 147 FR II galaxies, 1142 BL Lac objects, and 760 FSRQs, we consider the classification of their $\log L_{\text{kin}}$ values. To determine the classification boundaries, the GMM was applied to the distribution of kinetic jet power. The GMM assumes that the observed distribution can be described as a combination of several Gaussian components, each representing a potential subpopulation within the data. The BIC penalizes models with an excessive number of parameters and therefore helps to avoid overfitting. The optimal number of Gaussian components was quantitatively determined using the maximum-likelihood estimation algorithm (expectation maximization (EM)) together with the BIC for model selection. The analysis was performed in R statistical software. The package `mclust` implements an iterative EM algorithm to fit parameterized GMM and to perform maximum-likelihood clustering. There are two types of models fitted by `mclust`: the V model (univariate, unequal variance) and the E model (univariate, equal variance). Compared to the traditional Kaye's mixture mode (KMM) algorithm (K. M. Ashman et al. 1994), which is typically restricted to Gaussian components with equal variances, the GMM allows for more general cases, including unequal variance components, making it more suitable for real datasets. In addition, as shown in Figure 1(a), the BIC indicates that the V model is more optimal than the E model. The BIC values are shown in Figure 1(a). By comparing the BIC values for models with different numbers of components, the model that best describes the data without introducing unnecessary complexity can be identified. The model with the highest BIC value is generally considered the preferred choice.

In our analysis, we applied the BIC method to the $\log L_{\text{kin}}$ distribution of FR I, FR II, BL Lac, and FSRQ samples. The resulting BIC values were used to determine the optimal number of Gaussian components for the mixture model, which informs the classification of sources into distinct kinetic power clusters. The BIC values and the corresponding Gaussian densities are shown in Figures 1(a) and (b), respectively.

Table 3
Framework for Strong/Weak Jet Class

Population Characteristics	Strong	Weak
Definition	More powerful, single-component jet (single characteristic Γ) remains relativistic on large scales	Lower-power jet with faster deceleration, multiple emitting regions (multiple Γ)
Optical type	Mostly FSRQs	Mostly BL Lac objects
Synchrotron peak	All high-power LSP, some low-power LSP	All ISP HSP, some low-power LSP
Typical L_{kin}	$10^{43.5}-10^{46} \text{ erg s}^{-1}$	$10^{42}-10^{44.5} \text{ erg s}^{-1}$
Radio morphology	FR II; collimated jets, hot spots, edge-brightened	FR I; less collimated, no hot spots, edge-darkened
Accretion type	Efficient, likely standard disk	Inefficient, highly sub-Eddington

Note. The table summarizes the classification of AGN jets into strong/weak class proposed by E. T. Meyer et al. (2011). The categorization is based on multiple physical characteristics.

Although the BIC criterion formally favors a three-component GMM, the cluster 1 component only accounts for 2.3% of the total sample. This component corresponds to a low-weight Gaussian, which is introduced primarily to improve the overall likelihood of the model. Since the BIC involves a balance between model likelihood and complexity, even a Gaussian component with a very small mixing weight can be retained if it provides a statistically significant increase in likelihood. However, due to the extremely limited number of sources assigned to cluster 1, the parameters of this Gaussian component are poorly constrained. In particular, the estimated covariance matrix Σ becomes unstable, potentially leading to a severe underestimation of the intrinsic variance. As a consequence, both the mean vector μ and the decision boundary between this component and the neighboring component are statistically unreliable. For these reasons, cluster 1 was merged into cluster 2 in the final GMM classification. In Figure 1(b), the intersection of the two adjacent Gaussian curves occurs at $\log L_{\text{kin}} = 45.38$.

For comparison, when the model is restricted to two Gaussian components, the intersection gives a boundary at $\log L_{\text{kin}} = 45.83$. To further investigate this, we adopt a statistical approach. Noted by K. M. Ashman et al. (1994) and A. L. Muratov & O. Y. Gnedin (2010), the separation parameter (Δ) between two samples is calculated from their means μ_1 , μ_2 and standard deviations σ_1 , σ_2 : $\Delta \equiv \frac{|\mu_1 - \mu_2|}{[(\sigma_1^2 + \sigma_2^2)/2]^{1/2}}$. According to Δ , the separation of the two distributions is quantified. K. M. Ashman et al. (1994) suggested that $\Delta > 2$ is required for a clean separation. As shown in Figure 1(c), the model with two Gaussian components has a Δ value of 1.9 between cluster 1 and cluster 2. Similarly, in Figure 1(b), the model with three Gaussian components has a Δ value of 2.63 between cluster 2 and cluster 3. This result supports our preference for the three-component model over the two-component case. The GMM may decompose the data into multiple Gaussian components even when the distribution is intrinsically unimodal. Therefore, the BIC is adopted to determine the optimal number of components.

3.2. Comparison of Previous Work

The unified scheme for radio galaxies and blazars, in which FR I galaxies are the unbeamed counterparts of BL Lac objects and FR II galaxies are the unbeamed counterparts of FSRQs, was studied in the literature (P. Padovani 1992; G. Ghisellini et al. 1993; G. Z. Xie et al. 1993; C. M. Urry & P. Padovani 1995; M. Chiaberge et al. 2000). Based on Hubble Space Telescope observations of 110 BL Lac objects, C. M. Urry et al. (2002)

found that the host galaxies of these sources are predominantly luminous elliptical galaxies, showing a strong correspondence with FR I galaxies, while the hosts of high-luminosity quasars align with FR II galaxies. M. Giroletti et al. (2004) found that the intrinsic core radio powers of BL Lac objects and FR I galaxies are statistically similar. In the unified scenario for blazars and radio galaxies, HBLs and some LBLs may be unified with FR I LERGs as the same class. E. T. Meyer et al. (2011) proposed the concept of the blazar envelope. In the $\log \nu_{\text{peak}} - \log L_{\text{peak}}$ plane, they identified two distinct classes. The weak class, comprising IBLs, HBLs, and low-power LBLs, is expected to correspond to the FR I galaxies at smaller viewing angles. Conversely, the strong class, including FSRQs and high-power LBLs, corresponds to FR II galaxies. A summary of these jet classes is provided in Table 3.

In Figure 2(a), based on the framework of E. T. Meyer et al. (2011), the statistical distributions of the strong sources (blue) for FR II galaxies and FSRQs are largely consistent. However, FR I galaxies and HBLs exhibit a lower fraction of weak sources (red) than expected. This suggests that the framework may provide a broad overview. To refine the result for our sample, we adopt a new classification boundary $\log L_{\text{kin}} = 45.38$ derived from the GMM method.

We can derive three observations as follows. First, in this work, FR I galaxies show a clear correspondence with HBLs, with the weak sources being dominant. Similarly, FR II galaxies are well matched with FSRQs and are dominated by the strong sources. Specifically, IBLs are predominantly classified as weak sources, whereas LBLs exhibit an approximately equal distribution between weak/strong class. Overall, these results are broadly consistent with the framework described in Table 3.

Second, here we obtain an upper bound on the jet power for weak jets, $\log L_{\text{kin}} = 45.38$, which is consistent with the value, $\log L_{\text{kin}} = 45.27$ obtained by M. Keenan et al. (2021) using the scaling relation of K. W. Cavagnolo et al. (2010): $\log L_{\text{kin}} = 0.64(\log L_{\text{ext}} - 40) + 43.54$.

Third, in Figure 2(b), we find that 35 FR II galaxies and 99 FSRQs are classified into the weak class. This result is similar to the findings of M. Keenan et al. (2021), indicating that broad emission line sources do not always correspond to strong jet property. It is also in agreement with previous studies reporting the existence of low-power FR II galaxies (A. Capetti et al. 2017; B. Mingo et al. 2019). As noted by E. T. Meyer et al. (2011), optical spectrum type is not suggested to participate in the strong/weak classification, which relies instead on factors such as source morphology and core dominance. We adopt $\log L_{\text{kin}} = 45.38$ to link jet power with the accretion modes of weak and strong sources. The

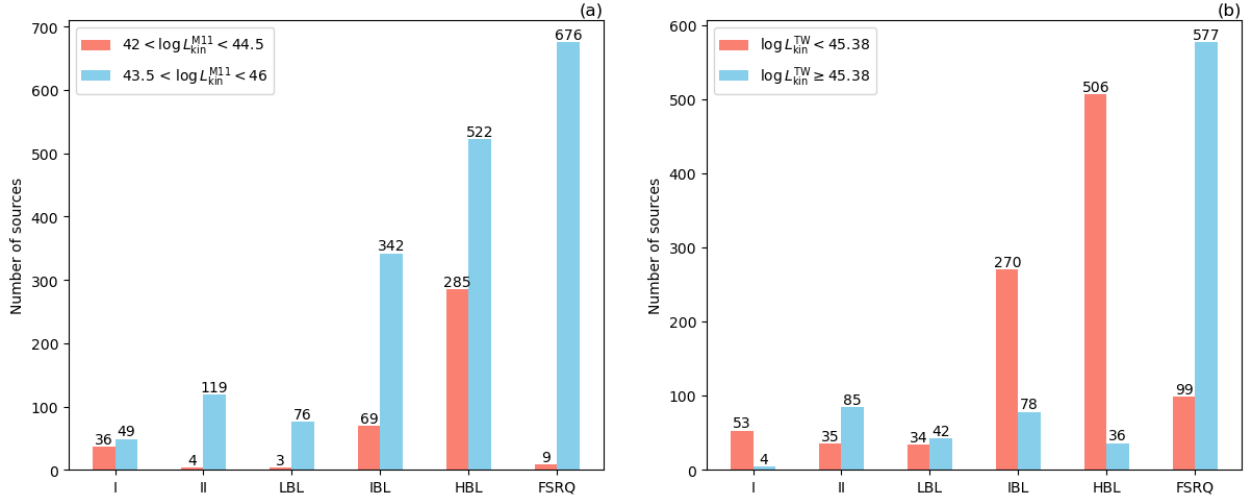


Figure 2. The histogram of strong and weak classes for each subsample. Red represents the weak class. Blue represents the strong class. (a) The distribution proposed by E. T. Meyer et al. (2011). 42 < log L_{kin}^{M11} < 44.5 represents the weak class. 43.5 < log L_{kin}^{M11} < 46 represents the strong class. (b) The boundary log L_{kin}^{TW} = 45.38 derived from this work.

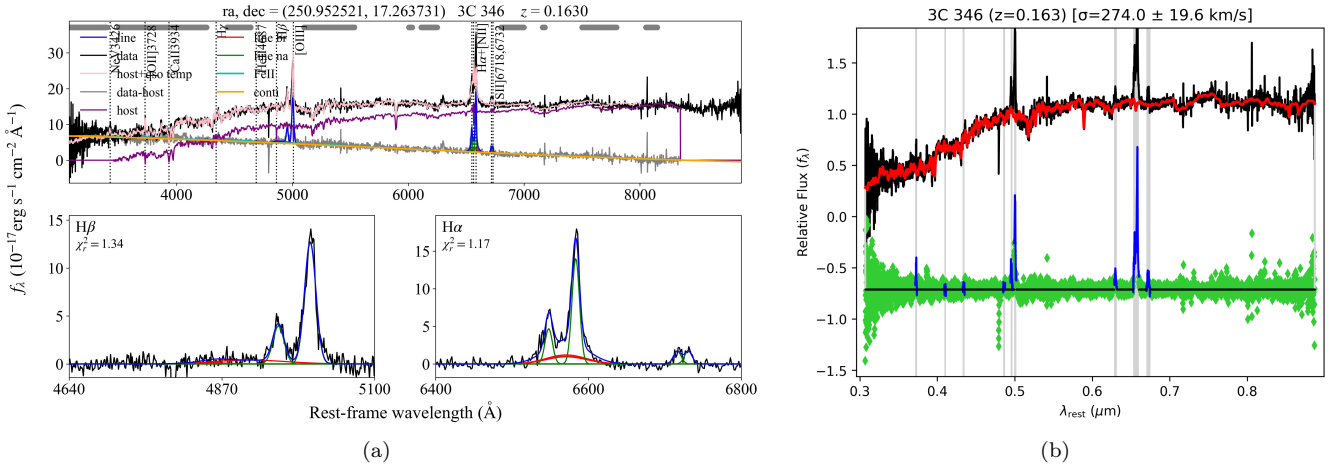


Figure 3. Spectral fitting example: 3C 346. (a) The QSOFITMORE fitting spectrum of 3C 346. The observed spectral data are shown as a black line. The broad and narrow components of the emission lines are represented by the red line and green line, respectively, while the modeled continuum is represented by the orange line. The blue line represents the sum of all components. (b) The pPXF fitting spectrum of 3C 346. The observed spectral data are shown as a black line, and the results of the stellar population synthesis are shown as a red line. The bottom panel displays the residuals of the fit.

analysis can be extended by incorporating the core dominance parameter, as discussed in Section 4.2.

4. Discussion

4.1. FR I–FR II and BLL–FSRQ Dichotomy

M. J. Ledlow & F. N. Owen (1996) provided important clues for the dichotomy division between FR I and FR II galaxies. Their results showed that the transition radio power between FR I and FR II galaxies is correlated with the optical magnitude of the host galaxy. Furthermore, the fundamental physical parameter governing the FR I/FR II dichotomy may be the Eddington-scaled mass accretion rate, which reflects the accretion mode and the properties of nuclear outflows (G. Ghisellini & A. Celotti 2001).

In this work, the kinetic jet power log L_{kin} is derived using Equation (1). In order to obtain the black hole mass data for the FR sample, we collect the black hole mass data for some

sources from the literature, while some others were calculated through the following process. We retrieve spectroscopic data from the Sloan Digital Sky Survey (SDSS) Data Release 18 (DR18). The Python package QSOFITMORE (Y. Fu 2021) uses χ² fitting to decompose the broad emission line and narrow emission line components, deriving the continuum luminosity (Figure 3(a)). According to the literature (Y. Shen et al. 2011; V. S. Paliya et al. 2021), the log M_{BH} can be estimated by Hα line FWHM and luminosity:

$$\log \left(\frac{M_{\text{BH}}}{M_{\odot}} \right) = 0.379 + 0.43 \log \left(\frac{L_{\text{H}\alpha}}{10^{42} \text{ erg s}^{-1}} \right) + 2.1 \log \left(\frac{\text{FWHM}_{\text{H}\alpha}}{\text{km s}^{-1}} \right), \quad \text{H}\alpha. \quad (4)$$

Similarly, following the literature (R. J. McLure & J. S. Dunlop 2004), the log M_{BH} estimation equations for Hβ

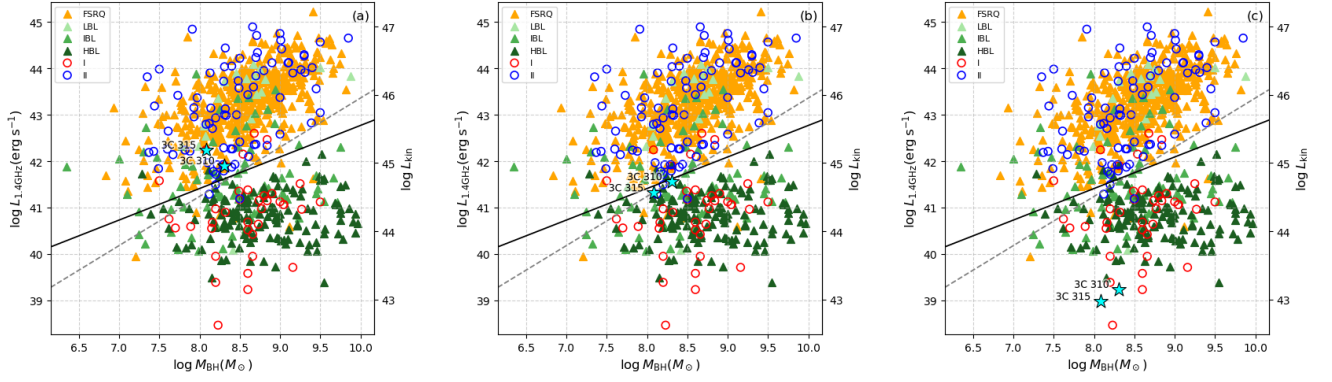


Figure 4. Black hole mass vs. 1.4 GHz luminosity diagram. Red open circles represent FR I galaxies, and blue open circles represent FR II galaxies. Light-green triangles represent LBLs, medium-green triangles represent IBLs, dark-green triangles represent HBLs, and orange triangles represent FSRQs. The black solid line indicates the dividing boundary obtained using a machine learning classification method. The gray dashed line represents the Ledlow–Owen dividing line between FR I and FR II galaxies. (a) The observed $\log L_{1.4\text{GHz}}^{\text{ob}}$ of 3C 310 and 3C 315. (b) $\log \delta = 0.08$ for 3C 310 and $\log \delta = 0.24$ for 3C 315. (c) $\log \delta = 0.6$ for 3C 310 and $\log \delta = 0.84$ for 3C 315.

and Mg II lines are as follows:

$$\log\left(\frac{M_{\text{BH}}}{M_{\odot}}\right) = 0.672 + 0.61 \log\left(\frac{\lambda L_{\lambda}}{10^{44} \text{erg s}^{-1}}\right) + 2 \log\left(\frac{\text{FWHM}_{\text{H}\beta}}{\text{km s}^{-1}}\right), \text{ H}\beta, \quad (5)$$

$$\log\left(\frac{M_{\text{BH}}}{M_{\odot}}\right) = 0.505 + 0.62 \log\left(\frac{\lambda L_{\lambda}}{10^{44} \text{erg s}^{-1}}\right) + 2 \log\left(\frac{\text{FWHM}_{\text{Mg II}}}{\text{km s}^{-1}}\right), \text{ Mg II}, \quad (6)$$

where λL_{λ} is the continuum luminosity at 5100 Å for H β and at 3000 Å for Mg II. We derived the stellar velocity dispersion (σ_*) using the Python package `pPXF` (M. Cappellari 2017). The stellar population model used for fitting was taken from GALAXEV (G. Bruzual & S. Charlot 2003). This program implements the penalized pixel-fitting method to extract kinematic information from galaxy absorption-line spectra. This program utilizes a maximum penalized likelihood approach to provide kinematic measurements (see Figure 3(b) as an example). Based on the obtained σ_* , we estimate the black hole mass using the $M_{\text{BH}}-\sigma_*$ relation from K. Gültekin et al. (2009):

$$\log\left(\frac{M_{\text{BH}}}{M_{\odot}}\right) = (8.12 \pm 0.08) + (4.24 \pm 0.41) \times \log\left(\frac{\sigma_*}{200 \text{ km s}^{-1}}\right). \quad (7)$$

The $\log M_{\text{BH}}$ data of FR galaxies are recorded in Table 1, column (12), while blazars are recorded in Table 2, column (11).

Figure 4 illustrates the relationship between $\log L_{1.4\text{GHz}}$ and $\log M_{\text{BH}}$ for the FR galaxies and blazars considered in this work, and the Ledlow–Owen dividing line is presented in the $\log M_{\text{BH}} - \log L_{1.4\text{GHz}}$ parameter space simultaneously. In this study, we applied the Support Vector Machine (SVM) method to search for the optimal separating boundary in the two-dimensional plane defined by $\log L_{1.4\text{GHz}}$ and $\log M_{\text{BH}}$. The SVM method is a common supervised learning algorithm

mainly applied to classification and regression analysis. Its objective is to find an optimal plane in the feature space that separates different classes as clearly as possible while maximizing the margin between them. In addition, we evaluated the reliability of the classification results using the average cross-validation accuracy. The resulting decision boundary can be expressed as

$$\log L_{1.4\text{GHz}} = 0.68 \log M_{\text{BH}} + 35.97. \quad (8)$$

The boundary achieves a training accuracy of 0.93 and a cross-validation accuracy of 0.93 ± 0.05 . The close agreement between the training and cross-validation accuracies indicates that the model does not show obvious signs of overfitting or underfitting. In Figure 4(a), six FR I galaxies (3C 28, 3C 29, 4C 20.20, 3C 310, 3C 315, and 3C 346) lie above the separating boundary, while two FR II galaxies (3C 88.0 and 3C 305) lie below it.

Based on radio morphology, J. P. Leahy & A. G. Williams (1984) classified 3C 310 as FR II with complex cross sections, while 3C 315 exhibiting extreme distortions was also classified as FR II. In the sample of F. N. Owen & R. A. Laing (1989), 3C 310 was described as a flat double source with bright outer rims of radio emission and roundish diffuse radio lobes. Because of its optical properties, it was classified as an FR I/II transition object. 3C 315 was identified as FR II, which is a classical source with compact outer hot spots and diffuse lobes. Combining radio morphology with optical properties, M. Chiaberge et al. (1999) reassessed both 3C 310 and 3C 315 as FR I. Due to the scarcity of changing-look FR galaxy candidates, it is difficult to determine the changing-look transition region. According to D. Huang et al. (2023), relativistic beaming can enhance the core emission of some FR I galaxies, causing FR I galaxies to appear as FR II galaxies. In Figure 4(a), 3C 310 and 3C 315 are highlighted with cyan stars, close to both the dividing line derived in this work and the Ledlow–Owen dividing line. Assuming that the intrinsic (or de-beamed) $\log L_{1.4\text{GHz}}^{\text{in}}$ of 3C 310 and 3C 315 just lie below the solid and dashed lines, we can estimate their Doppler factors. Under this assumption, the Doppler factors of 3C 310 ($\log \delta = 0.08$) and 3C 315 ($\log \delta = 0.24$) are estimated. Following the equation proposed by D. Huang et al. (2023), $\log \delta = (\log F_{1.4\text{GHz}}^{\text{ob}} + 2 \log d_L - 6.6)/(2 + \alpha)$, the Doppler factors of 3C 310 ($\log \delta = 0.6$) and 3C 315

($\log \delta = 0.84$) are obtained. In this case, our estimates are consistent with those reported by D. Huang et al. (2023). Adopting the intrinsic (or de-beamed) luminosity relation $\log L_{1.4\text{GHz}}^{\text{in}} = \log L_{1.4\text{GHz}}^{\text{ob}} - (p + 1)\log \delta$, the corresponding results are shown in Figure 4.

4.2. The Association between Core Dominance Parameter and Kinetic Jet Power

Relativistic beaming is an important phenomenon in the study of AGN properties. The unified scheme described by C. M. Urry & P. Padovani (1995) suggests that the classification of AGNs is fundamentally driven by orientation, with obscuration and beaming effect. The viewing angle θ is closely related to the Doppler factor δ . Various methods have been proposed to estimate the Doppler factor (J. Fan et al. 2014; G. Ghisellini et al. 2014; I. Liodakis et al. 2018). Determining the Doppler factor for FR radio galaxies is considerably challenging. D. Xiong et al. (2015) found that when the synchrotron peak frequencies of radio galaxies are not corrected for relativistic beaming, most blazars do not overlap with radio galaxies. They adopted the method proposed by T. Sbarrato et al. (2015), applied different bulk Lorentz factors ($\Gamma \sim 3, 5$, and 10) to radio galaxies, and investigated the connection with blazars. G. Giovannini et al. (1994) have shown that sources with larger core dominance correspond to smaller viewing angles θ and higher Doppler factors δ . In this work, beyond using $\log L_{\text{kin}}$, we further explore the unified scenario by incorporating the radio core dominance parameter $\log R_{\text{radio}}$ for classification.

The $\log R_{\text{radio}}$ for the blazar sample is taken from Z. Pei et al. (2020c; see Table 2). For the FR galaxies, VLBI data are applied to match the sources in our sample. The $\log R_{\text{radio}}$ values were derived using Equation (3), with the results summarized in Table 1. To verify the reliability of our calculations, the $\log R_{\text{radio}}$ results in this work can be compared with those of FR galaxy samples reported by Z. Pei et al. (2020a) and D. Huang et al. (2023) using the following statistical tests. The Kolmogorov–Smirnov (KS) test is a nonparametric statistical method utilized to examine whether the overall distributions of two samples are drawn from the same parent distribution. The Mann–Whitney U test is also a nonparametric test, providing a robust comparison of the central tendencies of two samples without assuming a normal distribution. The Student’s t -test is a parametric statistical test that evaluates whether the means of two samples differ significantly under the assumption of approximate normality. In these tests, the p -value is used to assess the statistical significance: $p < 0.05$ indicates a statistical difference between the two samples, while $p > 0.05$ suggests that no significant difference can be identified.

In the process of statistical testing, the validity of the KS test depends on the sample size n . In comparisons of two samples, the respective sizes (n_1 and n_2) should ensure statistical validity $\sqrt{(n_1 \times n_2)/(n_1 + n_2)} > 4$. The sample of this work is denoted as S_{TW} (with size n_{TW}). In the sample from Z. Pei et al. (2020a; S_{Pei}), $\sqrt{(n_{\text{TW}} \times n_{\text{Pei}})/(n_{\text{TW}} + n_{\text{Pei}})} = 3.87 < 4$ suggests that the p -value obtained by the KS test may cause certain biases. Specifically, the KS test returns $p = 0.03$, while the U test and the t -test result in p -values of 0.28 and 0.23, respectively. This discrepancy indicates that while there is no significant difference in mean values between S_{Pei} and S_{TW} , their overall distribution profiles differ. In contrast, for the sample of D. Huang et al.

Table 4
Statistical Test Results

Ref	n	λ_{KS}	KS Test	U Test	t -test
Z. Pei et al. (2020a)	21	3.87	0.03	0.28	0.23
D. Huang et al. (2023)	32	4.45	0.70	0.30	0.23

Note. Column (1): the references of previous studies. Column (2): the sample sizes of the compared groups. Column (3): λ_{KS} is the sample size validity defined as $\sqrt{(n_{\text{TW}} \times n_{\text{Ref}})/(n_{\text{TW}} + n_{\text{Ref}})}$. Columns (4)–(6): the p -values for the KS test, Mann–Whitney U test, and Student’s t -test, respectively.

(2023; S_{Huang}), $\sqrt{(n_{\text{TW}} \times n_{\text{Huang}})/(n_{\text{TW}} + n_{\text{Huang}})} = 4.45 > 4$. The KS test ($p = 0.70$), the U test ($p = 0.30$), and the t -test ($p = 0.23$) consistently show no significant difference between the $\log R_{\text{radio}}$ distributions of S_{Huang} and S_{TW} . The results are recorded in Table 4.

We plotted the sample in the $\log R_{\text{radio}} - \log L_{\text{kin}}$ plane (Figure 5(a)). When $\log R_{\text{radio}}$ is fixed, FR I galaxies transition into FR II galaxies, while HBLs transition into FSRQs as $\log L_{\text{kin}}$ increases. From a horizontal perspective, at a given $\log L_{\text{kin}}$, increasing $\log R_{\text{radio}}$ corresponds to the distribution FR I/HBL and FR II/FSRQ. This result is consistent with the unified scenario between FR galaxies and blazars. In Section 3.2, we noticed that once the dividing value of $\log L_{\text{kin}}$ changes, the weak/strong components of LBLs and IBLs vary significantly. Therefore, we attempt to use the SVM method in the $\log R_{\text{radio}} - \log L_{\text{kin}}$ plane to find the dividing line between the strong class and weak class. The SVM achieved a classification accuracy of 92%, and the resulting decision boundary is given by

$$\log L_{\text{kin}} = -0.02 \log R_{\text{radio}} + 45.16. \quad (9)$$

The classifier achieved a training accuracy of 0.921 and a cross-validation accuracy of 0.92 ± 0.02 . The close agreement between training and cross-validation accuracies indicates that the model shows no obvious signs of overfitting or underfitting.

To examine whether the above boundary equation can effectively separate the accretion rates of the samples, we calculated the Eddington ratio $\log(L_{\text{bol}}/L_{\text{Edd}})$ and divided the sources into two subclasses located above and below the SVM boundary line. The histogram distributions of $\log(L_{\text{bol}}/L_{\text{Edd}})$ for the two subclasses are shown in Figure 5(b). Two distinct Gaussian components can be clearly identified in the histogram. The strong class, located above the SVM line, exhibits high accretion rates (blue), whereas the weak class below the SVM line typically shows lower rates (red).

The core dominance parameter is often used as an indicator of orientation (e.g., F. Marin & R. Antonucci 2016). If the SVM boundary had a relatively steep slope, it might suggest that the beaming effect has an important role in the classification. However, the SVM boundary in the $\log R_{\text{radio}} - \log L_{\text{kin}}$ plane exhibits a relatively shallow slope, remaining close to a horizontal line at $\log L_{\text{kin}} = 45.38$. The work by M. Keenan et al. (2021) suggests that the range of the core dominance parameter depends on jet power, indicating that it is not purely orientation driven. In this work, kinetic jet power appears to be an effective parameter for classification in a statistical sense. The SVM boundary combines the intrinsic properties and orientation effects, but further investigation with complete samples is needed to better understand the

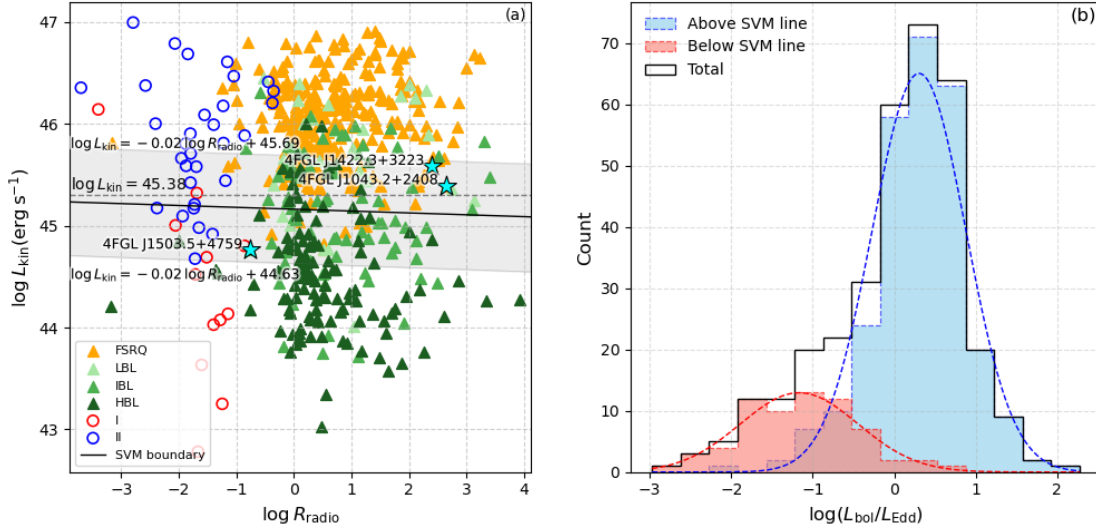


Figure 5. (a) The core dominance parameter vs. the kinetic jet power distributions. Red circles represent FR I, blue circles represent FR II, green triangles represent BL Lac (with color shading from light to dark indicating LBL, IBL, and HBL), and orange triangles represent FSRQ. The black solid line denotes the dividing boundary obtained using a machine learning method. The gray dashed line indicates the boundary $\log L_{\text{kin}} = 45.38$. The gray shadow area denotes the transition region, bounded by $\log L_{\text{kin}} = -0.02 \log R_{\text{radio}} + 44.63$ and $\log L_{\text{kin}} = -0.02 \log R_{\text{radio}} + 45.69$. Three changing-look AGNs are located in this region and are shown as cyan stars. (b) Accretion rate histogram. The red histogram represents sources below the SVM line, while the blue histogram represents sources above the SVM line. The black histogram denotes the entire sample.

interplay between intrinsic properties and environmental effects.

4.3. Changing-look Sources

The origin of the changing-look phenomenon is still under investigation, with explanations including intrinsic changes, temporary obscuration effects, and other possible mechanisms. Although some changing-look AGNs can be explained by obscuration, M. Sniegowska et al. (2020) find strong evidence for intrinsic changes. Dramatic variations in the accretion rate can cause the appearance or disappearance of broad emission lines. Some changing-look blazars exhibit variations in the equivalent width of broad emission lines, leading to transitions in classification between FSRQ and BL Lac.

In Figure 5(a), the SVM line in the $\log R_{\text{radio}} - \log L_{\text{kin}}$ plane suggests the possible existence of intermediate sources between the strong/weak class. Notably, some BL Lac objects and FSRQs exhibit overlapping regions, particularly in the vicinity of the SVM decision boundary. This statistical separation suggests the existence of intermediate or transitional sources between the strong class and weak class. In this work, we attempt to explore the distribution of changing-look blazars within this transitional region.

Bootstrap estimation is a sampling method with replacement adopted to quantify statistical uncertainties. Let the original dataset be denoted as $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, while N denotes the sample size. Bootstrap resampling is performed to obtain a set of SVM decision boundary parameters $w_{b,0}x + w_{b,1}y + b_b = 0$. Here $b = 1, \dots, B$ indexes the bootstrap realizations, where B is the total number of resampling iterations. $w_{b,1}$ and $w_{b,2}$ are the coefficients of the SVM decision hyperplane corresponding to x and y . b_b is the intercept term of the SVM decision boundary for the b th bootstrap realization. The mean value and standard deviation are subsequently calculated: $\bar{y}(x) = \langle y_b(x) \rangle$, $\sigma_y(x) = \sqrt{\langle (y_b(x) - \bar{y}(x))^2 \rangle}$.

The 3σ interval is adopted as the transitional region, thus mitigating the effects of sample variance.

The gray shadow area in Figure 5(a) illustrates the transition region held between two lines $\log L_{\text{kin}} = -0.02 \log R_{\text{radio}} + 45.69$ and $\log L_{\text{kin}} = -0.02 \log R_{\text{radio}} + 44.63$. The sources 4FGL J1043.2+2408 ($\log L_{\text{kin}} = 45.39$), 4FGL J1422.3+3223 ($\log L_{\text{kin}} = 45.59$) and 4FGL J1503.5+4759 ($\log L_{\text{kin}} = 44.76$) are located within this region and are marked with a cyan star. H. A. Peña-Herazo et al. (2021) classified 4FGL J1043.2+2408 as a BL Lac object and 4FGL J1503.5+4759 as an FSRQ based on an optical analysis of Large Sky Area Multi-object Fiber Spectroscopic Telescope (LAMOST) spectra. In contrast, in this work, 4FGL J1043.2+2408 is classified as an FSRQ, while 4FGL J1503.5+4759 is identified as an IBL. H. D. Mishra et al. (2021) reported that 4FGL J1422.3+3223 undergoes multiple transitions between BL Lac and FSRQ states. In the study by Z. Pei et al. (2022), these sources were distributed within an “appareling zone” on the plot of broad-line region luminosity versus γ -ray luminosity and classified as changing-look blazars.

The interpretation of the changing-look phenomenon remains broad. The brightness of the beamed emission can overwhelm the spectral lines. If the source is misaligned, the spectral lines would emerge (e.g., C. M. Urry & P. Padovani 1995; P. Giommi et al. 2012). D. Garofalo et al. (2010) proposed a framework involving mixed black hole spin states. HERG FR II galaxies with retrograde spin and efficient accretion can spin down to a prograde configuration, evolving into radiatively inefficient FR I galaxies. However, if the black hole spin transition occurs before or after the change in accretion efficiency, the source may exhibit a hybrid state. D. Huang et al. (2023) discussed the changing-look phenomenon in FR galaxies. Some FR I galaxies might have weak radiative cooling owing to the beaming effect, while emission from the core is enhanced. As a result, their broad emission lines can be overwhelmed by the nonthermal jet continuum, causing FR I galaxies to appear as FR II galaxies. In previous studies (e.g., E. T. Meyer et al. 2011; M. Keenan et al.

2021), transitional populations are often absent, either appearing at lower power or missing because of selection effects. The states observed in changing-look sources suggest the presence of transitional populations. In this section, some changing-look sources transitioning between BL Lac and FSRQ states are found to lie close to the SVM boundary in the $\log L_{\text{kin}} - \log R_{\text{radio}}$ plane. This suggests that such sources may occupy a transitional region between different accretion modes. The appearance of changing-look AGNs in this transitional region provides observational support for a more dynamic view of the AGN unified scenario. We expect that, as larger samples become available and the physical nature of changing-look AGNs is better understood, this will help improve our understanding of the AGN unified scenario and galaxy evolution.

5. Conclusion

In this work, we compile a sample consisting of 75 FR I galaxies, 147 FR II galaxies, 1142 BL Lac objects, and 760 FSRQs to investigate the unified scenario of FR galaxies and blazars. Based on the radio luminosity at 1.4 GHz, we derive the kinetic jet power (T. M. Heckman & P. N. Best 2014). We apply BIC and GMM approaches to identify a boundary for separating different subclasses. In addition, the SVM method was utilized to derive a dividing line in the $\log R_{\text{radio}} - \log L_{\text{kin}}$ plane. Our main results can be summarized as follows:

1. The intersection of the two Gaussian components at $\log L_{\text{kin}} = 45.38$ is adopted as the boundary value between the strong class and the weak class.
2. Combining FR galaxies with blazars, a machine learning linear relation was obtained: $\log L_{1.4\text{GHz}} = 0.68 \log M_{\text{BH}} + 35.97$. This relation is consistent with the Ledlow–Owen dividing line (M. J. Ledlow & F. N. Owen 1996) and supports a dichotomy in the unified scenario.
3. We also define an SVM boundary in the $\log R_{\text{radio}} - \log L_{\text{kin}}$ plane. The resulting relation is $\log L_{\text{kin}} = -0.02 \log R_{\text{radio}} + 45.16$. Sources above this line have higher accretion rates, while those below it have lower accretion rates.
4. We proposed a changing-look transitional region held by $\log L_{\text{kin}} = -0.02 \log R_{\text{radio}} + 44.63$ and $\log L_{\text{kin}} = -0.02 \log R_{\text{radio}} + 45.69$ in the $\log R_{\text{radio}} - \log L_{\text{kin}}$ plane; sources in that region are changing-look AGN candidates.

Acknowledgments

We are grateful to the anonymous reviewer for the constructive suggestions and comments. We also thank Zhiyuan Pei and Xuhong Ye for helpful discussions on the properties of the radio core. This research is partially supported by the National Natural Science Foundation of China (NSFC grant 12433004), the National Key Research and Development Program of China (grant No. 2025YFA1614102), the Eighteenth Regular Meeting Exchange Project of The Scientific and Technological Cooperation Committee between the People's Republic of China and the Republic of Bulgaria (Series No. 1802), and grants from the China Manned Space Project (grant No. CMS-CSST-2025-A07). Additional support was provided for the Astrophysics Key Disciplines of Guangdong Province and Guangzhou City, as well as the Guangdong Major Project of

Basic and Applied Basic Research (grant No. 2024A1515013169). This work is also supported by the Key Laboratory for Astronomical Observation and Technology of Guangzhou. This research has made use of the NASA/IPAC Extragalactic Database, which is funded by the National Aeronautics and Space Administration and operated by the California Institute of Technology.

ORCID iDs

Jingyi Ma  <https://orcid.org/0009-0007-9118-8326>

Junhui Fan  <https://orcid.org/0000-0002-5929-0968>

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